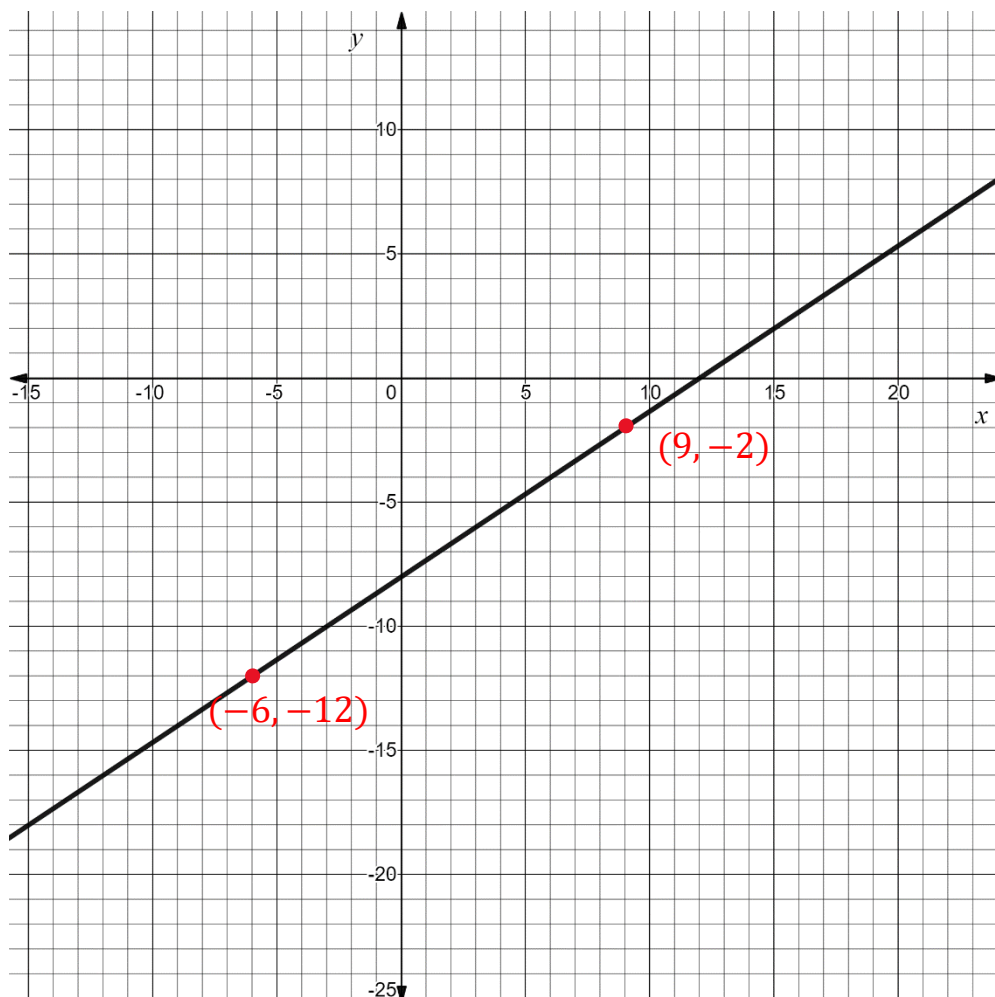


Algebra 1
Unit 4
Lesson 7 Homework

Name **Answer Key** _____
Date _____
Period _____

The function $f(x) = \frac{2}{3}x - 8$ is shown on the coordinate grid.



- Use the graph to find the solution $f(x) = -2$. Label the point that represents the solution.

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- Solve $\frac{2}{3}x - 8 = -2$. Show your work.

$$\begin{array}{r} +8 \quad +8 \\ \cancel{\frac{3}{2}} \cdot \cancel{\frac{2}{3}} x = \frac{3}{1} \cdot \frac{3}{\cancel{2}1} \end{array}$$

$$x = 9$$

- Use the graph to find the solution $f(x) = -12$. Label the point that represents the solution.

$(-6, -12)$

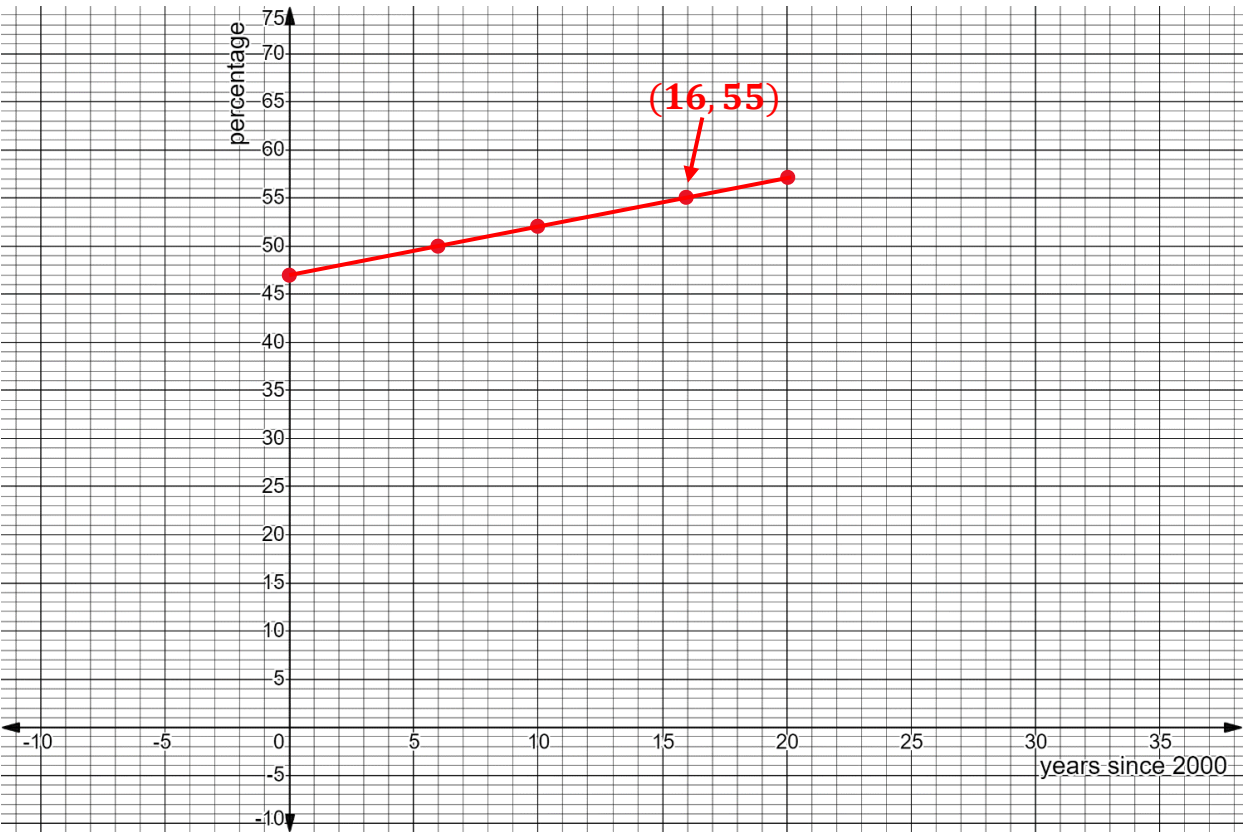
- Solve $\frac{2}{3}x - 8 = -12$. Show your work.

$$\begin{array}{r} +8 \quad +8 \\ \cancel{\frac{3}{2}} \cdot \cancel{\frac{2}{3}} x = \frac{2}{1} \cdot \frac{3}{\cancel{2}1} \end{array}$$

$$x = -6$$

From 2000 to 2020, a world health organization collected data on the percentage of a country’s population that had access to safe drinking water. The function $W(t) = 0.5t + 47$ models the percentage of the population in Guatemala that has safe drinking water where t is the number of years since 2000 and W is the percentage of the population that has access.

- 5. Explain why the values of t should have a constraint.
The values of t should have a constraint because the data was only measured for 20 years (2000 to 2020).
- 6. Write the constraint for the values of t using set builder notation.
 $\{x|0 \leq x \leq 20\}$
- 7. Write the constraint for the values of W using set builder notation.
 $\{w|47 \leq w \leq 57\}$
- 8. Graph the function using the constraints from questions 6 and 7.



- 9. Use the graph to find the solution $W(t) = 55$. Label the point that represents the solution.
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- 10. Solve $0.5t + 47 = 55$. Show your work.
$$\begin{array}{r} 0.5t + 47 = 55 \\ -47 \quad -47 \\ \hline 0.5t = 8 \\ \frac{0.5t}{0.5} = \frac{8}{0.5} \\ t = 16 \end{array}$$